



Market bifurcations in board sports: How consumers shape markets through boundary work

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ABSTRACT

The market-shaping literature recognizes that consumers are actors who can shape markets. However, research into the mechanisms of consumption-driven market-shaping is only emerging. This paper shows that one way in which consumers shape markets is through boundary work, as consumers make, break, and re-make the boundaries among multiple markets. Empirically, this paper investigates how user innovation practices catalyzed the formation of four board sport markets: surfing, kitesurfing, windsurfing, and stand-up-paddling. The findings show that consumers shape markets in three steps: (1) generating local variations of consumption practices; (2) imposing order and coherence to intermediate frames; and (3) channeling creative consumption practices into market formation. This paper contributes to the market-shaping literature by conceptualizing how consumers splinter an existing singular market into new markets. We coin the term ‘market bifurcations’ to describe how a singular, seemingly cohesive consumer practice breaks down, splintering into several local variations, which can then catalyze the formation of new markets.

1. Introduction

Market-shaping is an empirical phenomenon in which market actors transform their business landscape rather than only reacting to it (Maciel & Fischer, 2020; Nenonen, Storbacka, & Windahl, 2019). For scholars investigating market-shaping, markets are complex adaptive systems that can be the subject of innovation (Araujo, 2007; Kjellberg, Azimont, & Reid, 2015; Nenonen et al., 2014). Examples of market-shaping acts include the creation of derivatives in financial markets (MacKenzie, 2006), reconfiguring petrol stations into service stations (Azimont & Araujo, 2007), and normalizing gambling from illegal to legitimate (Humphreys, 2010).

As the literature on market-shaping grows, we learn more about the different market-shaping actors (Kindström, Ottosson, & Carlborg, 2018), which include not only firms (Baker & Nenonen, 2020) and regulators (Depeyre & Dumez, 2009), but also consumers (Giesler & Fischer, 2017; Hietanen & Rokka, 2015; Martin & Schouten, 2014). Consumers shape markets in multiple ways (Harrison & Kjellberg, 2016), including through acts of creative consumption in which consumers generate new music genres (Dolbec, 2015), sports (Diaz Ruiz, Peñaloza, & Holmqvist, 2020), or hobbies (Martin & Schouten, 2014). Consumers shape markets by challenging online exchange practices for music (Giesler, 2008), disintermediating market actors like the music

producer (Hietanen & Rokka, 2015), or developing infrastructure for a new sports category (Martin & Schouten, 2014), among other examples. However, while consumers can generate effects in markets, it remains unclear how consumption practices intertwine with market-shaping. To be precise, current studies lack a comprehensive conceptualization of how consumers shape markets.

This paper addresses the research gap by investigating one specific mechanism through which consumers shape markets – *boundary work* (Star & Griesemer, 1989). This term emphasizes the work required for making, maintaining, and redressing the demarcations in-between multiple markets, plural (Chimenti, 2020; Ellis, Jack, Hopkinson, & O'Reilly, 2010). Indeed, one premise of market-shaping is the existence of various markets (Diaz Ruiz, 2013; Mele, Pels, & Storbacka, 2015), each one featuring distinct yet overlapping organizing practices (Kjellberg & Helgesson, 2007). However, distinguishing markets from one another is challenging (Kjellberg & Olson, 2017). As Frankel (2018, p.461) writes, “singling out a particular market as an object of inquiry raises not least questions of what is implied by ‘market’ in this particular situation and to whom it is a market.” If Frankel (2018) is correct, then one limitation of the market-shaping literature is that the construction and maintenance of market boundaries are often invoked rather than analyzed, especially when consumers participate. In consequence, we ask: *How do consumers use boundary work to shape markets?*

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This paper investigates market-shaping through the creative consumption and user innovation (Branstad & Solem, 2020; von Hippel, 2005) that have re-constructed the boundaries of four sports markets often grouped together as *boardsports* – surfing, kitesurfing, stand-up-paddling (SUP), and windsurfing. Consumers have played a prominent role in developing and commercializing boardsports gear. In a study of 57 equipment innovations, Shah (2000, p.3) wrote, “innovations in skateboarding, snowboarding, and windsurfing equipment were typically developed by a few early expert participants in those sports, lead users, and also by some of those same lead users after they founded small companies to produce their innovations for sale.” Therefore, especially in sports markets, consumers often shift roles from users to instructors, innovators, and entrepreneurs (e.g., Diaz Ruiz, Peñaloza, et al., 2020).

As creative consumption blurs the divide between consumption and market practices, it constitutes boundary work – i.e., the process of re-making demarcations in-between distinct and distinguishable fields. We argue that consumers participate in boundary work by developing local variations that are never entirely new, as they borrow from adjacent fields to form a new singular activity. This paper shows how consumers re-arrange inputs into new configurations that become sufficiently distinct and distinguishable to form new markets.

Empirically, the paper investigates boundary work in intertwining boardsports by analyzing their narrative histories, or ‘biographies’ (Spring & Araujo, 2017), through the perspective of consumers who are lead users (Branstad & Solem, 2020) and gear tinkerers (Martin & Schouten, 2014). We inquire into consumer journeys – from amateur to industry participant – to provide a glimpse of the ‘origin stories’ that stabilize each boardsport as a distinct yet distinguishable field. Hence, this paper conceptualizes how consumers engage in boundary work to set up and maintain distinctions in-between adjacent markets.

2. Literature review

Building upon *Constructivist Market Studies* (Araujo, 2007; Kjellberg & Helgesson, 2007) and *Market System Dynamics* (Nøjgaard & Bajde, 2020) perspectives, the market-shaping literature (Nenonen et al., 2019) challenges notions of the market as one single entity (market-as-noun), focusing instead on the underlying organizing practices (market-as-verb) (Mele et al., 2015). This transition emphasizes the “doings” of markets (Kjellberg & Helgesson, 2007), thus characterizing markets as unfolding or continuously in the making (Kjellberg et al., 2015). Multiple actors participate in the formation of market norms, exchanges, and representations; they include not only firms, consumers, and regulators, but also non-human ‘actants’ such as technologies, discourses, and devices (Çalışkan & Callon, 2010). In the market-shaping approach (Nenonen et al., 2019, p.252), markets are “complex adaptive socio-technical-material systems, consisting of institutions, actors, practices, and discourses that organize particular economized exchanges.”

Market-shaping is an empirical act in which one or several actors attempt to proactively transform a complex market system for their benefit (Nenonen et al., 2019). Some examples include market shaping in the steel industry (Kindström et al., 2018), the introduction of free-range eggs that incorporate concerns for animal welfare into what is otherwise considered a commodity (Diaz Ruiz, Baker, Mason, & Tierney, 2020), and the reconceptualization of how a circus operates (Baker, Storbacka, & Brodie, 2019).

2.1. How consumers shape markets?

The current literature recognizes that consumers can shape markets (Harrison & Kjellberg, 2016; Martin & Schouten, 2014); however, the specific mechanisms remain vague. For example, after reviewing previous research in which consumers have shaped markets (Hietanen & Rokka, 2015; Scaraboto & Fischer, 2013), Baker et al. (2019, p.304) concluded, “This sociological view recognizes that (1) micro-level

practices collectively shape a market, and (2) as practices change so does the market and vice versa.” However, it is not certain that micro-level consumer practices change market practices (or vice versa). The mechanisms of how micro-level consumer practices can lead to market formation remain unclear.

The most common understanding of the consumers’ role is as “potential demand” (Harrison & Kjellberg, 2016). In the user innovation literature (von Hippel, 2005), consumers’ familiarity with products and services constitutes a source of ‘democratic’ innovation that firms can use to develop commercial products. Thus, user innovation represents a proof-of-concept developed by the potential buyers themselves. However, according to Harrison and Kjellberg (2016, p.448), this approach both “exaggerates and underplays” the capacity of consumers to shape markets.

Consumers also shape markets through activism (Den Hond & De Bakker, 2007). One example is Scaraboto and Fischer (2013) investigation of how *fatshionistas*, plus-sized consumers vested in fashion trends, directed their efforts to influence fashion designers to adopt an inclusive approach to body types. In another example, Rosenthal and Cardoso (2016) investigated how Brazilian football fans boycotted FIFA’s (Fédération Internationale de Football Association) mega-event (FIFA World Cup) as a protest.

A third way in which consumers shape markets is through creative consumption. In this type of consumption, dispersed yet collaborative consumer efforts generate new consumption practices that *could* lead to market formation (Giesler & Fischer, 2017). For instance, Martin and Schouten (2014) investigated how consumers tinkered with minimotos in their backyards in ways that catalyzed a new market. As the minis rose in popularity, they became the anchor for a market made up of dedicated events, media outlets, distinctive products, and infrastructure built-for-purpose.

Creative consumption can shape markets through playful collaboration among consumer collectivities (Biraghi, Gambetti, & Pace, 2018). For example, Kozinets (2007) documents how Star Trek fans wrote fan fiction and produced fan-made films that kept the Star Trek franchise active for decades. Similarly, Goulding and Saren (2007) discuss how consumer groups organized Goth-themed festivals, thus becoming entrepreneurial tribes. Moreover, consumers often start firms building upon their focal activity (Branstad & Solem, 2020), and during this process, consumers can steer market practices.

The influence that consumers exert in markets is salient in the music industry. For example, consumers have challenged record companies by extracting music from compact discs (CD) and sharing it via peer-to-peer platforms (Giesler, 2008), thus reframing what constitutes the object of exchange. Hietanen and Rokka (2015) describe how enthusiasts in the dubstep scene distributed their music through counter-cultural channels. By doing so, consumers disintermediated the music producer and transformed distribution channels to make “virtually every owner of a personal computer [is] a potential music producer” (Hietanen & Rokka, 2015, p.1571).

The emergence of variations of cultural products within a market is the subject of recent inquiry. For example, Coskun-Balli and Ertimur (2017) investigated variations of yoga practice within the U.S. yoga market. Their investigation highlighted the divergence of yoga practices over 30 years, resulting in partially overlapping yet distinct cultural products, variously emphasizing spiritual, medical, fitness, or commercial meanings. However, for Coskun-Balli and Ertimur (2017), these yoga variations are all part of a single “American yoga market,” rather than multiple, distinct markets, plural.

The previous examples show that consumers’ creative consumption practices can be a source of market innovation (Biraghi et al., 2018). However, while we know that creative consumption generates variations of consumption practices, it is unclear how it shapes markets. In this regard, previous work hints at the importance of local innovation. Dolbec (2015) proposed the notion of ‘local innovation networks’ to conceptualize how consumers work together to create variations of

cultural products, such as Electronic Dance Music (EDM). To form a market, local entrepreneurs commercialized EDM music first as niches with multiple local variations across sites, then expanded into the mainstream music market.

Building upon Dolbec (2015), we distinguish three levels in which creative consumption is generative of market formation. First, user innovations generate multiple cultural products that result in local variations. Examples include the variations between salsa dancing styles in Cuba, New York, and Los Angeles (Diaz Ruiz, Peñaloza, et al., 2020). Second, actors impose order by organizing the variations. For example, Scott, Cayla, and Cova (2017) explain the emergence of commercial obstacle races featuring pain and mud. Third, mainstream actors identify (and co-opt) niches when the cultural product is stable enough to become part of a market. For instance, while streaming was previously a niche, market actors have now integrated online streaming as the new mainstream (Disney +, Apple TV +, Amazon Prime).

As consumers generate new local variations, the resulting arrangements can blur distinctions between markets. Kjellberg and Olson (2017) coined the term ‘adjacent markets’ to conceptualize partially overlapping markets. Their study of two cannabis markets reveals different market practices, depending on their recreational or medicinal purposes. The theoretical problem, however, is how do we distinguish among multiple adjacent markets? To that end, we turn to the literature on boundary work.

2.2. Re-defining market boundaries: The role of boundary work and objects

To (re)define the boundaries of a specific market, observers often rely on abstractions, representations, and categorizations (Diaz Ruiz, 2013; Diaz Ruiz & Kjellberg, 2020) to argue that a collection of some practices (but not all) constitutes a single market. To analyze the process of delineating distinctions between market representations, we draw upon the notion of *boundary work* (Chimenti, 2020; Ellis et al., 2010; Frankel, 2018). This notion refers to sustained efforts towards achieving and maintaining independent, stable, concrete, and settled concepts, in this case defining one singular market, by setting up boundaries.

For Ellis et al. (2010, p.228), boundary work means unpacking the elements “which are taken into account and those which are ignored” when it comes to determining what counts as a market, and from whose perspective. The central idea behind boundary work is the recognition that most concepts we consider stable, concrete, or settled have required sustained efforts to achieve such perceived stability.

The episteme of boundary work emerged from the analysis of demarcations and divisions between fields of academic knowledge, specifically how these divisions are created, reinforced, and enacted. Recent work in constructive market studies has used the notion of boundary work to investigate the formation of market boundaries (Chimenti, 2020; Ellis et al., 2010; Frankel, 2018). However, an epistemic challenge is demarcating boundaries that are porous and prone to change. Finch and Geiger (2010) addressed this challenge through boundary objects, defined as “abstract or concrete objects, whose structure is sufficiently common to several social worlds to ensure minimum identity in terms of the intersection while being sufficiently flexible to adapt to the specific needs and constraints of each of these worlds” (Trompette & Vinck, 2009, p.8). Thus, boundary objects are robust enough to maintain identity across disciplines, yet plastic enough to admit multiple interpretations.

Star and Griesemer (1989) coined the notion ‘boundary objects,’ whether material or abstract, to conceptualize crossings over disciplinary boundaries that enable shared understandings (Carlile, 2002). One example is how scientists and bureaucrats share databases and classifications that allow for swift coordination without fully understanding each other’s disciplines (Spree & Jarzabkowski, 2009). In Star and Griesemer (1989) work, the coordination mechanisms between scientists and bureaucrats in a natural history museum included

classifications, conventions, and standards that inhabit both scientific and administrative fields.

For Carlile (2002), boundary objects are an essential part of boundary work because they help emphasize dependencies across fields, especially by delineating how to achieve practical solutions to shared problems. For example, Humphreys (2010) documents the construction of a shared understanding of gambling in the media in ways that legitimized the practice. Casino owners and other stakeholders normalized what otherwise were illegal activities by shifting narratives and semantic categories to describe gambling. By doing so, gambling became legitimate, while simultaneously creating distinctions from other adjacent practices such as illegal betting. Kjellberg and Olson (2017) research on cannabis markets shows the boundaries of these markets emerge through legislation, that is, understanding the adjacent markets as either medical or recreational. In this paper, we focus on boundary objects to study how consumers build demarcations between markets, but also how they trespass seemingly stable boundaries.

While the literature on boundary work focuses on demarcations, it does not explain how closely intertwined fields relate to one another. We also know little about how boundary work creates distinctions between adjacent markets. To be precise, how does a group of actors construct demarcations that feature resemblances and also emphasize distinctions in-between existing and new markets?

Returning to Dolbec (2015) EDM, and Hietanen and Rokka (2015) Dubstep, if we conceptualize the music market as the core, then both Dubstep and EDM would form part of a belt of peripheral variations sharing counterculture as a boundary object. We can draw a similar interpretation of Coskun-Balli and Ertimur (2017) American yoga and Shah’s (2000, 2006) boardsports, namely that consumer practices are never entirely new, but emerge from local variations. Therefore, while user innovations may result in entirely separate markets, this outcome is not inevitable. Instead, user innovations may generate a new consumer practice without becoming a market. To address this gap, the rest of the paper empirically investigates how user innovations demarcate the market boundaries of boardsports.

3. Method

3.1. Investigating boundary work through origin narratives

To address the research question (how do consumers use boundary work to shape markets?), we explored the formation of boundary objects operating across multiple boardsports. The notion of boundary objects emerged from the field of Science and Technology Studies (STS) (Star & Griesemer, 1989). Its material-semiotic approach has carried through into the field of market studies (Araujo, Kjellberg, & Finch, 2010; Finch & Geiger, 2010). Building upon this semiotic-material approach, Spring and Araujo (2017) used ‘product biographies’ to investigate how “products are chronically unstable, both physically and institutionally,” thus requiring managerial efforts to stabilize them, albeit temporarily.

By focusing on narrative biographies of boardsports, we aimed to uncover the unfolding practices that have come to define their markets. Hence, we first paid attention to the narratives of how each boardsport emerged – their origin stories – and second, accounts of how each boardsport relates to and deviates from other boardsports.

We chose boardsports because their multiple variations have become popular as separate markets (pumpedupsup.com, 2020). Industry reports reflect the popularity of boardsports, with estimates of 5–23 million surfers worldwide (International Surfing Association, n.d.). In 2011, 1.2 million people were SUP boarding (Channel Signal, n.d.), and in 2012 there were 1.5 million kitesurfers (ISAF report). These figures indicate that even almost a decade ago, millions of people worldwide were taking part in boardsports.

All four water boardsports are hybrids in that enthusiasts tend to

cross over (e.g., windsurfers Neil Pryde and Robby Naish have both designed innovative kiteboards), and thus object variations emerge across boardsports. The most notable hybrid device is the board itself, which is essential in all four sports, but with variations. Furthermore, boardsport origin stories often feature a heritage of ancient Polynesian practices, with contemporary innovations involving a board variation in combination with another device (e.g., sail or kite).

3.2. The fieldwork

We started by consulting secondary sources to learn the origins of each boardsport, including how boardsports differ from one another. A plethora of popular online publications explain the technical differences between the boardsports (e.g., blog posts such as www.kitesurfer.com/kitesurfing-vs-surfing). Manufacturers publish similar material in online articles (Boardsport Source, 2016, 2019), and there are discussions between self-appointed experts in online forums. Additionally, scholarly publications in the orthopedics, sports, and engineering literature offer glimpses into the technical differences between boardsports (Shah, 2006; Schram, Hing, & Climstein, 2016; Van Bergen, Commandeur, Weber, Haverkamp, & Breederveld, 2016). We used these descriptions to identify differences and similarities between boardsports in the gear, boards, skills, and locations.

However, while popular publications were a fruitful starting point, they were insufficient to understand how consumers create and maintain distinctions among the activities. Therefore, we built upon previous work on the surfing community by Akaka and Schau (2019) to inquire about the journey from newcomer to expert. Further, aware that Shah (2000) identified lead users as key drivers of innovation in boardsports, we conducted fieldwork with consumers who had transitioned into some form of commercial operation. These individuals know the commercial aspects of the sport due to selling equipment, instructing users, and participating actively in community events. We collected 13 semi-structured in-depth interviews with respondents who were simultaneously avid boardsports consumers and commercial operators, including retailers, owner-operators, entrepreneurs, instructors, and photographers (Table 1). Participant selection was achieved through snowball sampling and purposive sampling, where the researchers directly approached retailers and boardsports magazine editors in New Zealand (N.Z.) identified through secondary data such as newspaper articles.

Interviews lasted around 90 min, and we asked about three topics: first, their personal journey; second, origin stories of their preferred boardsport practice; and third, their involvement in the commercial side of the sport. We started by asking general questions about boardsports, such as “How does each boardsport differ?”, and then progressed to questions about their journey with their boardsport(s), and a description of their market. Because we were using a semi-structured approach, the interviewees were able to drift the conversation onto their preferred topics. We discussed changes in their practice relating to physical aspects such as gear, cultural aspects including social norms and cultural meanings, and (as a surprise to us) current innovations (e.g., foiling). It was a surprise because we expected commercial operators involved in SUP, for example, would want to stick to the topic of paddleboarding, but our interviewees were keen to discuss current innovations emerging in adjacent markets. The interviews were recorded and transcribed verbatim, amounting to 384 pages.

While the interviews formed the core of the fieldwork, several interviewees referred to current events, for instance applications to the International Olympic Committee, or made references to popular culture, especially documentaries. Therefore, we expanded the fieldwork to include references from popular culture mentioned by our interviewees. Since we were looking for origin stories, documentaries such as Dogtown and Z-Boys (2001) further enhanced the narratives of how a particular sport came to be at a specific point in time. We also included additional sources, which were not directly referenced by the

interviewees (Table 1). The criteria for inclusion were that sources directly address the narrative origins of the boardsports and their evolving sets of qualities as they adapt continuously to moments of transition and instability. The fieldwork also included 42 pages of reflective fieldnotes jointly taken by the researchers. These multiple sources provided a rich dataset from which we were able to draw robust conclusions.

3.3. Data analysis

To explore how consumers use boundary work to shape markets in four boardsport activities – surfing, windsurfing, kitesurfing/kiteboarding, and SUP, we were interested in consumer perspectives on their multiple beginnings. In the analysis, we focused on the work and effort required to achieve the currently perceived stability of the boardsports, albeit temporary (Spring & Araujo, 2017). We therefore analyzed the unfolding socio-technical arrangements (Diaz Ruiz, Peñaloza, et al., 2020). We also investigated why each boardsport is an oddly similar variation instead of a replica. The analysis generated understandings of the local variations, and how these variations catalyze the formation of adjacent markets.

Useful for our analysis is DeLanda (2016) notion of ‘bifurcations,’ which emphasizes the transitory nature of the singular, its collapse, and the emergence of the multiple. Using bifurcations as an analytical lens means identifying the periods of transition in which what we thought of as stable becomes unstable, generating episodes of uncertainty and producing alternatives. Hence, we paid close attention to the devices, natural objects, spaces, and brands as they form a transitory unit, which is subject to multiple ways of failing. We inquired after breakdowns, failures, and misalignments that catalyzed periods of transition and variation. Hence, we analyzed the data not as historical accounts, but as competing stories with multiple origins.

These narratives should not be confused with factual history. Their storytelling is instead akin to origin stories intended to make sense of complex events at times of intensive change. Interviewees connected past and present causally through stories that explain how each boardsport began to be. The stories emphasize change rather than stability, stressing the unfolding qualities that develop into local variations. By placing attention on the narrative connections, we identify boundary work as the performance of a fraught singularity susceptible to bifurcations (Table 2).

4. Findings

The boardsports enthusiasts – turned market actors – gifted us with narratives of the many beginnings of each boardsport. Their accounts demonstrate that the boundaries separating the boardsports are porous, dynamic, and have multiple overlaps in terms of both technical solutions and social categorization. To be precise, while Shah (2000, 2006) demonstrated that consumer participation is essential for gear innovation in boardsports, here we explain why this context is such a fertile ground for user innovation (Branstad & Solem, 2020). Consumers grow frustrated with the lack of ‘board time’ in the water due to marine weather conditions, i.e., too much or little wind, and the absence or presence of waves. Reminiscent of the *frustrated fashionistas* documented by Scaraboto and Fischer (2013), frustrated boardsports consumers aim to change their sports by reimagining how to increase their board time. Some consumers then develop commercial practices to form first niche- and then mainstream markets. We introduce below an overview of how creative consumption catalyzes the emergence of multiple markets, and then explain each step in detail in the following sections.

Once consumers start to generate variations, their work catalyzes market emergence in three steps (Fig. 1). First, consumers generate multiple cultural products by tinkering with gear and practices. Consumers innovate by putting together salient features across fields in a

Table 1
Fieldwork: Primary and secondary data sources.

Primary data sources: Interviews		
Pseudonym	Occupational background	Board consumption expertise
Robin	'Garage' entrepreneur – hemp materials designer	Surf, skateboards, snowboards
Lynda	Hemp materials designer (surf lifestyle clothing)	Surf, skateboards, snowboards
Vivi	Retail owner-operator: clothing, accessories	SUP
Nik	Retail owner-operator: clothing, accessories/ SUP instructor	SUP
Hailey	Retail store branch manager	Surf, skateboards, SUP, bodyboards
Greg	Photographer, magazine editor, book publisher	Surf
Alex	Windsurf school instructor/ owner-operator	Windsurf, SUP, kitesurf
Robbie	Retail brand ambassador, SUP instructor, experiential leader	SUP
Darrel	Retail store marketing manager	Surf, bodyboards, skateboards
Dr Ollie	Board and endurance trainer for race competitors	Surf, snowboard, skateboards, wakeboards
Jim	'Garage' entrepreneur – designs/builds boards	Windsurf/foil, SUP racing, surf, kiteboards, skateboards
Joshua	'Garage' entrepreneur –board designer/ amateur SUP racer	SUP, surf, kiteboards, skateboards, skimboards
Andreas	Retail owner-operator: clothing, accessories wholesaler, surf instructor, board repairs	Surf, skateboards
Secondary data sources		
Category	Definition	Curated Selection
Boardsport history	Scholarly research on contemporary history/orthopedic/sports	• 'The World in the Curl: An unconventional History of Surfing,' Journal of Sports Sciences
Catalogues from manufacturers	Accounts of design, raw materials, innovation of different boards	• Websites/in-store catalogues: 'Robby Naish,' 'Neil Pryde,' 'Starboards,' 'Slingshot,' 'Red Paddle Co.' • Australian wholesaler 'Shapers Manufacturer Co.,' N.Z. custom board shaper 'Pete Anderson.'
Magazines/newspaper	Accounts of board uses and interviews with producers/consumers	• Mpora Magazine(e.g., 'Robby Naish interview: The man who conquered the ocean one competition at a time' • The Guardian (e.g., 'A swell century: surfing Sydney's beaches from 1915 to 2019 in pictures')
Online forums/blogs	Consumer, manufacturer, sport tourism companies, travel bloggers, competitors/racers publishing opinion pieces	• SustainableSurf.org (e.g., 'Is It Possible to "Wipe Out" the CO2 Footprint of Your Surf Travel?') • Forum.Surfer.com (Surfboard Design)
Films/documentaries	Accounts of board cultures and interviews with consumers/entrepreneurs	• The Endless Summer; Cinema V (1966) • Dogtown and Z-Boys; Sony Pictures Classics (2001)

'combinatorial' type of innovation (Yoo, Boland, Lyytinen, & Majchrzak, 2012). Second, some consumers venture into commercial practices that stabilize the variations into local niches, popularizing their use among other consumers. While we find some commercial elements at this stage, the niche is the catalyst of market emergence, is not yet a mainstream market. Third, fueled by their increasingly popular consumption practices, consumers-turned-entrepreneurs generate the market infrastructure and engage in legitimizing efforts, e.g., entering the Olympics, participating in self-regulation, and drafting industry-level standards. Fig. 1 shows a non-linear progression from consumption to markets due to the active involvement of user-innovators in stabilizing cultural variations, first as niches and then into markets.

4.1. Consumers generate local variations

While most consumers practice surfing or SUP conventionally, a minority engage in a generative form of consumption that co-creates new uses and meanings. Previous consumer research includes ample evidence that, especially in highly interactive contexts such as sporting activities (Martin & Schouten, 2014), consumers' work is generative, for instance when using consumption to define and transform a cultural production and sense of identity. One example is how individual consumers integrate the surfer's ethos into their worldview (Akaka & Schau, 2019; Canniford & Karababa, 2013). In this fieldwork too our interviewees frequently mentioned a surfing culture. For example, married surf entrepreneurs Robin and Lynda described surfers "left-leaning" political ideology, sustainability values, and 'hippy' lifestyle.

Historically, surfers have been associated with the protest movement; they're more left-leaning, liberal, kind of anti-nuclear... because they're

already like outcasts being more comfortable and a bit 'more hippy.'
There's definitely a sustainability push from surfing. – Robin and Lynda

Consumers' cultural production often involves the formation of consumer collectives. Previous research by Canniford and Shankar (2013) shows the pursuit of ideas of purity in nature through group dynamics is an integral part of the surfing culture. Vivi, a certified SUP instructor and shop owner, explained that people who surf often adopt an ethos visible through consumption, and which can last a lifetime.

People with a surf culture, they've started surfing from a young age and they've probably progressed through. [...] They got into surfing, then shortboard surfing because they've become more advanced. As they get older they get a little slower, a bit bigger and arthritis kicks in or something, so they go for the bigger, longer boards [...] but they've generally surfed most of their lives and have grown with a surfboard at some point whereas I find it different with the paddleboard industry that they're coming in [without the surf culture]. – Vivi

While previous work refers to the consumer culture as singular, studies in consumer research show that consumers generate multiple local variations. For instance, Diaz Ruiz, Peñaloza, et al. (2020) found that consumer groups manifest over multiple consumption sites through similar (but not identical) variations in techniques, locations, and gear that can generate distinct and distinguishable versions. Another example is Coskuner-Balli and Ertimur (2017) finding that different variations of American yoga emphasize spiritual, medical, fitness, or commercial meanings. In support of this stream of research, our empirical data show that N.Z. surfing has developed its own culture, which is oddly similar yet distinct from other surfing traditions, especially that of Australia. For example, in the quote below, Greg, author of several surf books and editor and photo director of a surfing magazine,

Table 2
Emerging themes & table of quotes.

Theme	Sub-topic	Exemplar quotes
Consumers generating cultural products	Individual identity projects	“When I grew up, surfing – and surf culture – was an escape, from school, from parents, from mainstream culture. The technologies were simple; a board, wax, and someone’s elder brother who had a car big enough to haul longboards to the beach. The excitement of catching a big wave was tempered with the (sometimes forced) serenity of waiting for a decent set to come in.” - Wendy
	Collective identities	“Surfing is very much a connected community and you end up with friends, often called [a tribe] ... and it got penned a long time ago, since the 1970 s, it was an international tribe. There’s very much a huge amount of interconnectivity between surfers worldwide, for sure.” - Greg
	Identifiable consumer culture	This documentary film [Dogtown and Z-Boys] depicts the surfing and skateboarding subcultures from California in the 1970s. Dogtown, a depressed lower-middle class area in Santa Monica witnessed the birth of contemporary skateboarding when disenfranchised teens unable to surf all day turned to skateboarding in abandoned swimming pools to emulate the rush of surfing waves. - Stacy Peralta (documentary)
User-innovators impose coherence into niche markets	Creative consumption	“I used to build all my windsurfs back in the 1980 s, like from my very first one. It was a fail, but I started from there and it’s only recently that I’ve started buying windsurfing boards. There’s 40, 50 or 60 h if you want to be realistic in building a board. So it’s a hobby, not a money-saving device.” - Jim
	Tinkering variations	“Wakeboarding used to be called ‘skirting.’ The idea of that was started by a couple of N.Z. surfers of what to do when it’s flat. Well, let’s get towed by a boat [...] to get going on our boards.” - Greg
	Stabilizing the variations into niche markets	“We add things. So Joshua [son/business partner] and I have got skateboards but we add a windsurfing rig to it, and we’ll go windsurfing. People come up and say: Where’d you get that from? Did you do it yourself? It’s super-cool. It’s almost as good as windsurfing. So we got a skateboard, mounted a windsurfing universal joint on the front of it, and then put a small windsurf rig on it. You can do it for five minutes because there’s no set-up. The other thing is set-up time. It’s a new type of skating really.” - Jim
Niches evolve into ‘adjacent’ mainstream markets	User innovation	“There’s a story that if you put any foil into any old board, you’re going to rip your fin box out. BUT some of the ones we use have got this little plate... the vertical part of the foil is called the mast. At the top of the mast there’s this plate. It goes across the bottom of the board. So, it spreads the load. Because we use foils like that, we don’t have trouble so we can use our foils into any board we want.” - Alan
	Enlisting coalition of market allies	“There was a law change [Advertising Standards Authority in N.Z.] where in advertising you can’t... claim an unquantifiable thing. But with that law change, the only reason that [boardsport companies] would ever come into question is if someone brings it into question. No one’s going to go and get prosecuted unless there’s a complaint. I think what we’re experiencing at the moment is those retail companies [claiming market leadership] are trying to fly under the radar of making unsubstantial claims and getting away with it.” - Greg
	Distinguishing against adjacent practices	Product Review: “The trend for 2019 is this Wing-Surfer – an inflatable wing, which is a cross between a kiteboarding sail and a windsurfing rig, but with no strings or rigging needed. It allows a very simple, pure method for a board rider to harness the wind, be it on a foiling kiteboard or a SUP – just blow it up, grab the handles and go. The Wing-Surfer was developed by legendary Hawaiian windsurfer Robby Naish and is built like a kiteboarding sail using a Quad-Tex canopy and Teijin Dacron leading edge.” - Yachting World Magazine

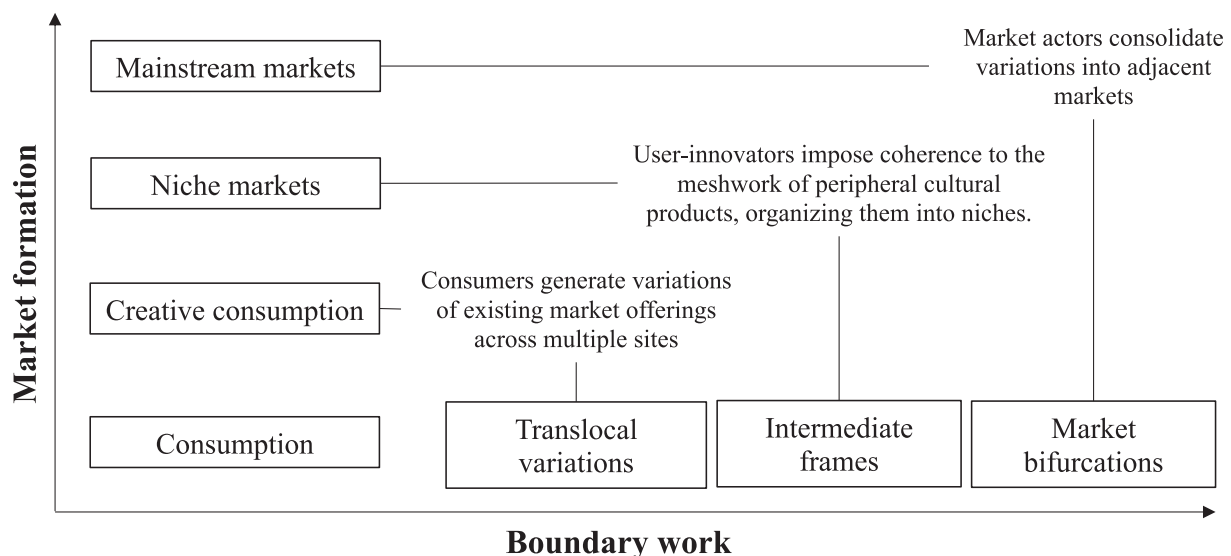


Fig. 1. From consumption to markets: How consumers use boundary work to shape markets.

describes how the local surfing community in N.Z. builds its group identity by emphasizing the importance of the Australian surfing community while maintaining distinctness within N.Z.

... very much in N.Z. in the 1970s, was a strong thing in the surfing community, that kind of rite of passage to spend some time in Australia back then. So as a young lad over that next year I was able to save up enough money to go. [...] I think surfing culture, especially with surfing culture in Australia and N.Z., rides on playing each other out. – Greg

Previous studies emphasize in-group social dynamics to explain cultural shifts. For instance, Schau, Muñiz, and Arnould (2009) studied social and discursive group dynamics based on sharing goals, meanings, and practices. To participate in consumption communities, consumers learn the rituals and norms. However, the emergence of cultural products is not due to sociality alone. While our data also show the importance of in-group sociality, a new finding is the effect of local or situational impositions, especially maritime conditions, which impinge upon how consumers engage in boardsports. For instance, Hailey, a national retail store branch manager and avid surfer described the conditions for kitesurfing as contradictory to the conditions for surfing: “Kitesurfing’s only good when it’s onshore [wind that blows from a large body of water onto a landmass] and the waves are terrible [for surfing] and that’s probably where some surfers may [try kitesurfing] but I don’t know many.”

The empirical data in this study reveal the importance of maritime environmental constraints in limiting how much ‘board time’ consumers manage to get. Each boardsport requires and depends upon local environmental conditions, especially the presence of waves and wind. Certain boardsports are only possible under the right conditions, but not all. For example, below Lynda explains that each boardsport requires a different understanding of the meteorological and geographical constraints. She first distinguishes between ‘wave-based’ and ‘wind-based’ boardsports, and then goes into more detail about the qualities of the waves – “really punchy, close-up kind of waves” for bodyboarding, while kitesurfing works on the “back of the surf [waves].” With the addition of a hydrofoil (a device extending below the board into the water, designed to lift the board off the surface to create speed), the practitioner can go “foiling... far out” once the foil starts to generate lift.

Bodyboarding and surfing are ‘wave-based.’ You need good conditions to perform those activities. Bodyboarding’s quite unique because you’re surfing so close to the wave and in a short space you can surf quite unique waves, some really punchy, close-up kind of waves. It’s harder for surfers to stand up because the waves are really fast and just dumping, it’s harder to pop up whereas with bodyboarding you can just shoot. So, kitesurfing and windsurfing are ‘wind-based.’ For SUP you can just go out when there’s nothing. Then there are the crossovers, especially kitesurfing where they can use their kite to go out the back of the surf [waves] and combine the two. They’ve got kitesurfing surfboards, I’ve seen a few. Then foiling has made a big leap combining them because you’ve got foiling SUP boards and they can catch waves very far out – once it generates lift there’s a minimal kind of resistance so that’s kind of combining it again. – Lynda

The geography of N.Z. supports multiple boardsports, with each location enabling or limiting different versions. The west coast and east coast offer very different conditions. Vivi explained that the popularity of SUP in N.Z. is partly due to the presence of sheltered bays and tides that at times result in “flat water” without waves or current. While these geographical constraints (e.g., sheltered bays featuring flat water) are anathema to kitesurfing or windsurfing, they create playgrounds for paddleboarders. Flat waters, according to Vivi, are essential for learning how to paddleboard.

This local area in N.Z. has really safe coastlines for learning how to paddleboard. We’ve got sheltered bays. If someone’s wanting to start,

we’ll take them down to Orewa Estuary where it’s sheltered... take them at high tide so there’s no current... flat water so they’re not feeling unbalanced with the chop and it’s smooth, so people start paddling. We teach them how to paddle, the technique... it’s like riding a bike, once you’ve got the balance sorted through your body you don’t need to concentrate on balance anymore and you just focus on your paddle technique. – Vivi

As local conditions vary, consumers incorporate new ways to respond and adapt to the shifting conditions. For instance, Robin, a self-described “garage entrepreneur” interested in a variety of boardsports, explained that the rapidly changing meteorological conditions in N.Z. – either too much or too little wind – motivate consumers to shift into a wider range of water-related sports and gear. Robin described the emergence of a local type of consumer called “watermen” who can pick and choose from gear for multiple boardsports to fit the weather.

If you had all the different equipment you could go out any day of the week, depending on the conditions which in N.Z. is so variable. You’ve got beautiful calm days. You’ve got crazy stormy days, everything in between from both coasts. I think that’s where a lot of Kiwis, these “watermen,” have all these different crafts they can choose to use whichever day suits the best. – Robin

As local impositions foster divergence across sites, the baseline consumption practice no longer applies. Consumers develop new uses and meanings in ways not recommended, intended, or even imagined by manufacturers. As consumers adapt their practice to multiple sites, creating local variations, they form a dispersed meshwork of barely coherent consumption practices. These variations are *translocal* in that each variation makes sense in the local context. However, when analyzed from a meso-level market perspective, the variations result in a meshwork of fringe, ambiguous, and contradictory consumer activities with nascent yet limited influence. The following sections explain how the work of user-innovators and other consumption-driven entrepreneurs contributes to simplifying (some of) the variations, stabilizing them into (more or less) coherent niche markets.

4.2. Imposing local coherence via niche markets

We know that when consumers shape markets, they mobilize multiple activities, including activism (Scaraboto & Fischer, 2013), or overlapping industry functions (Hietanen & Rokka, 2015). Accordingly, this section explores how consumers use creative consumption to shape markets by imposing order in commercial market niches.

We find that N.Z. boardsports enthusiasts tinker with their gear in ways that traverse the boundaries between consumers and producers, participating in user innovation as “garage entrepreneurs” who initiate commercial ventures servicing their fellow enthusiasts. All the interviewees were both avid sport enthusiasts and industry participants. In some cases their transition from consumer to industry participant emerged as an organic progression to fund their passion, as Robin and Lynda explain below.

Robin: I’ve got a mate shaping some boards [manufactures surfboards]. He surfs as well.

Interviewer: What got him into shaping boards?

Robin: He loves surfing. He has a huge love for surfing, and I guess [he] wanted a job or be able to be as close to [surfing] as he could.

Lynda: I know that he recently went traveling, and in each surf town he would shape boards to fund his travels and find mentors to help develop his shaping. He’s been gone for a year almost.

The story of the transition from consumers to commercial operators and even inventors is one of passionate engagement. When we inquired about how the interviewees got into the business side, they all pointed first to their profound immersion in their sports activity, love of the ocean, and aspiration for a lifestyle in which they can earn money doing

what they love; the commercial opportunities emerged later. For example, Vivi and her husband Nik became avid paddleboarders before they opened their shop. When it comes to gear innovation, the boundaries between consumption and professional work are blurry. For Jim and Joshua, tinkering with boards is at the core of both these activities to the extent that no distinction can be made between consumption and production; gear innovation is both.

When I was in my mid-teens, 16–17, windsurfing was just taking off in N.Z. I pestered my dad, bought some foam, bought some fiberglass, and built my first one [windsurf board], which lasted for about two hours on the water before it fell to bits. But yeah, that's the sort of original story that all kids have when spending time at the beach. – Jim.

In line with previous theorization by Shah (2000, 2006), board-sports enthusiasts are keen innovators, tinkering constantly with their gear. Why then is the boardsports context such a fertile ground for user-led innovation? After analyzing the interviews, we place the answer in frustration. To explain further, consumers are frustrated about not having enough 'board time' due to local environmental impositions forming roadblocks and obstacles to their enjoyment of their sport. Precisely, the frustration arises from waiting for the right conditions to surf, which leads to the consumer quest to be 'out there' in any meteorological condition. Frustration is consistent with previous research by Scaraboto and Fischer (2013), who describe how fashion consumers use activism to shape markets in order to become active participants in the fashion field. When it comes to frustration, Greg explained that the basic conundrum that surfers experience is a lack of waves.

There's very much a finite number of waves. At any beach the waves come in a set. In that set there's talk of seven waves. It's not seven... more like five. When a set comes in there are five waves to catch and you want to be on one of those five waves. If there are 30 people in the water competing for those five waves, it's a problem. – Greg

The driver for gear innovation in the boardsports context boils down to a combination of frustration avoidance and thrill seeking. Our interviewees variously described responses to the exasperation of dealing with local impositions (e.g., the weather) that cause downtimes: "They came up with foiling because they were annoyed that they couldn't ride all the time" (Vivi); "They [invented windsurfing] because they were tired of waiting for waves" (Alan); or "It was too windy [to surf], so they found out how to use the pull from the kite" (Hailey). The other side of frustration is thrill-seeking. Consumers innovate to experience the thrill of riding faster. As Alan discussed, this driver has popularized the hydro-foiling innovation because it is faster than any other boardsport: "It's growing fast because you are going fast [...] it's like you are surfing the air."

The process of user-driven innovation revealed in the data is 'combinatorial' (Yoo et al., 2012), whereby consumers incorporate features from multiple sports into new combinations and variations. These variations become hybrids, occupying spaces in-between the porous boundaries of boardsports. To stabilize the ragtag collection of hybrids, tinkering consumers coin and circulate a range of labels and neologisms designed to explain where the innovation originated. Combining features under new labels stabilizes the new gear by hinting at the familiarity of well-established practices. Below, self-proclaimed garage entrepreneur and board shaper Jim describes putting together gear from multiple sports to make new arrangements that they then designate through hybrid labels. The labels for these new arrangements have the advantage of identifying the salient features that make them recognizable: the sail, kite, or paddle. Different to the boundary objects that identify features the variation has in common with another boardsport, we call the distinguishing salient features of each consumption variation *signature objects*.

It's kind of a natural flow of taking a kit from one boardsport to another. It's trivial in a way, but when people see it, they notice how straightforward it is. Because they know it's just a mix of two things, and that's

the best innovation – I know that [gear] and I know that [gear]; let's put them together [...] I mean this isn't a secret. The 'stickskating' where you've got to paddle or 'paddleskating,' or 'landboarding'... where you're stroking with a paddle-like stick. It's just taking the legwork out of it and giving you a bit more board time. It feels engineering but it feels like sort of a natural toolmaker. It's a toolmaker sensation where you're going - Oh, if I could do that, that would be better - so you can fashion something up, check out if it works or not and when it does work, I mean it's pretty satisfying. – Jim

The generative entrepreneurial work in which consumers work as innovators is not an extension of their consumption activity; instead, gear-innovation is their hobby. As consumers tinker with their devices as part of an activity they are passionate about, sometimes they produce market-shaping innovations. For example, our interviewees referred to Simon Anderson, an Australian competitive surfer and inventor credited with inventing the three-fin surfboard, "the thruster", in the 1980s. The three-fin feature became an industry standard for surfboards.

Simon Anderson's quite famous in the surf industry. He invented the 'Thruster' which was new to surfing. He's done surfboards and stuff like that. Now he's got into paddleboards. Then he stopped doing paddleboards probably because they were a lot more expensive but there are also issues sometimes with the construction of them. They [paddleboards] were so big, [and with] new technology their team went their separate ways. He's gone back to his market - surfboards. – Vivi

While consumers mix features from parallel sports, this customization is still a consumption practice, not a market practice. The difference is that market practices must include valuation and exchange (Kjellberg & Helgesson, 2007). Hence, we emphasize the role of consumers in stabilizing the meshwork of variations into identifiable niche markets (see Fig. 1). Consumers distill "a complex jumble of otherwise ambiguous and contradictory activities, pronouncements, and impressions, into a simplified and relatively coherent portrait" (Ashforth & Humphrey, 1997, p.53).

[Robbie Naish] has a new 'batwing' coming out that you can just hold on to and foil at the same time. They're developing that so it's almost like bringing kitesurfing into foiling, which is done already but this time it's a kite that you hold onto rather than being connected to barring lines... these people are trying to push the limits all the time and then they learn from that. – Vivi

In the interviews, the boundaries between consumption and production blur as consumers double as commercial operators. Moreover, their innovation moves through market boundaries. Through novel combinations, consumers bring together multiple features and functions from across a range of sports; the result is a new boardsport variation. The end-game is more board time and being 'out there' in the action. And yet, as Martin and Schouten (2014) notes, while customization practices can catalyze new markets, they are not themselves markets; at least not until commercial exchanges stabilize. Therefore, while this section shows that consumers can produce innovations, the resulting variations remain dispersed in translocal consumption variations. The question remains, how does innovation in consumer practices catalyze market emergence?

4.3. Niches splinter into 'adjacent' mainstream markets

Popularity generates pressure to standardize gear, processes and exchange practices, coalescing them into market-like practice. To do this, actors recognize the importance of stabilizing what would otherwise be a set of diffuse and ultimately wilting consumer efforts. Most of the previous work on market-shaping in the existing literature describes this legitimizing stage (Dolbec & Fischer, 2015). For instance, Baker et al. (2019) discuss the disruption of the old circus based on exotification and animal cruelty, and the emergence of a new circus based on

mesmerizing performances and talent à la Cirque du Soleil. In our empirical data, we find evidence that the process of legitimizing consumer variations into a market requires standardization, coalitions of market allies (c.f., Baker & Nenonen, 2020), and building distinctions/boundaries against well-established practices.

As noted in the dataset, multiple boardsports including surfing, paddleboarding, windsurfing, and kitesurfing have existed since antiquity. Archeological evidence shows that for 3,000 years fishermen in what is now Peru have used a reed watercraft called ‘caballitos de to-tora’ for both fishing and recreational ends (Johnstone, 1988). This traditional practice resembles paddleboarding in that an elongated reed shaft like a kayak paddle is used. Surfing also has ancient roots in ethnic communities in the South Pacific, and its contemporary leisure variation can be traced back at least to Hawaii in 1885 (Nendel, 2009). Therefore, why is SUP boarding or surfing’s commercial popularity only emerging now? Previously, only communities living near by the appropriate maritime conditions could surf. However, recent innovations are making boardsports more popular by making the boards transportable, cheaper, safer, and easier to use. One example is inflatable paddleboards that require less room to store and can be reduced to fit into conventional luggage for airline travel. Since carrying and storing windsurfing gear is cumbersome, making a board that can be deflated contributes to the removal of entry barriers, and thus popularization. According to SUP boards retailer ‘Red Paddle’, inflatable boards decrease the barriers of entry, accessibility, and skill level.

I’ve seen windsurfing go through the guises and what has become a very extreme and specialist sport. It kind of forgot the every man and woman on the beach wants to have fun. Paddle Co. offers access to the water in a nontechnical way. Before Paddle Co., boards were big, heavy and hard to transport and store. I used to paddle with friends who loved it but would readily admit they would never buy one as they were simply too cumbersome, [...] I immediately thought that if I could solve the performance issues affiliated with inflatable technology, then iSUPs would become the most relevant product for most people. – John Hibbard, InflatableBoarder.com

With popularization, the new problem of compatibility emerges; the gear from one brand is incompatible with other brands. In the water and at high speeds, incompatible gear can lead to severe injuries. Combining gear from multiple boardsports also elevates the risk of accidents, to the extent that kitesurfing was once classified a dangerous, extreme sport. For the survival of kitesurfing, multiple producers agreed to cooperate in setting up industry standards. In the quote below, Jim discusses how establishing kitesurfing involved a concerted effort to reduce the level of risk through collaboration between multiple industry leaders.

My understanding is that 12–15 years ago, the whole [kitesurfing] industry [came together]. There were so many major injuries and deaths in this new sport that all the major industry players got together, went into a retreat and agreed they were going to unify the gear, and standardize stuff because it was just so difficult for consumers [...] If they [consumers] picked up someone else’s gear that was different it could be lethal, so they agreed [on standardization] for the sake of the industry [...] they made it easier, and they invested a lot more in product design. [...] Every year it’s getting better and better. There are marked changes in hardware even from last year to this year. – Jim

Normative practices are part of market formation (Kjellberg & Helgesson, 2007). In our data, participation in boardsports is discussed in terms of navigating the regulatory norms and compliance requirements. For instance, consumers often transitioned into the industry by capitalizing on their expertise to become professional instructors (Diaz Ruiz, Peñaloza, et al., 2020). However, as Robbie mentions below, becoming a professional instructor and tour operator in N.Z. involves compliance with existing regulation on ‘adventure’ activities. This regulatory burden nudged Robbie to take practical decisions in his

paddleboarding practice, such as not venturing more than 100 m from the safety of land.

In order to be a SUP instructor there’s a couple of things. You need to have the right concession - you need to be allowed on certain land/water. That’s permission from council authorities. Second thing is the correct safety management system. In the outdoors we call it an SMS [Safety Management System] which is part of your SOPs [Standard Operating Procedure]. Now paddleboard instruction may or may not be a regulated activity depending on a few things. For example, if you go further than 100 m into sea [...] not even to sea, if you’re 100 m from safe land, then it becomes ‘adventure-regulated.’ I’ve deliberately been offering a level of SUP activity that’s not regulated because I don’t want to have the compliance cost of being regulated. – Robbie

The most widely used approach to market-shaping in the literature is to mobilize institutional lenses to explore how one stigmatized activity becomes a legitimate market. For example, Humphreys (2010) studied how gambling became a legitimate (and legal) market. In our fieldwork, participants also mentioned the importance of legitimizing what would otherwise remain garage inventions and marginal customization practices. One specific strategy for gaining legitimacy is becoming an Olympic sport. Today, kitesurfing and kiteboarding are among the fastest growing water sports. According to Business Wire (2018), the kitesurfing/kiteboarding market is projected to reach around USD \$300 million by 2022, fueled in part by the sports incorporation into the Olympics. This challenging process requires regional and international tournaments, regulations, and a consolidated fan base. For instance, windsurfing became an Olympic sport for men in 1984 and for women in 1992, but the first windsurfing world championship was held decades earlier in 1973 (Everything Windsurfing, 2018). Below, Alan raises the likelihood that foiling will be included in the Olympics. He explains that part of foiling’s projected growth is the large number of places available to practice the new sport, as it does not require specific types of waves, just wind.

Originally it [windsurfing] was huge. Windsurfing has been an Olympic sport since the late 1980s and still is [popular]. But now they’re looking for 2024 to put ‘windfoiling’ in the Olympics. ‘Kitefoiling’ will also enter in the 2024. ‘Foiling’ is really growing. [...] How many places in the world have waves? [...] it is easier to find countries with winds. – Alan

Once the popular appeal of the variation grows, a second requirement is defining the variation. From an industry perspective, the challenge is to classify or categorize the hybrid as part of the taxonomy the industry already has, but in ways that explain both the similarities and differences. One example is describing the combination, e.g., partly kitesurfing, but with a foil. The hybrid forges distinctiveness through boundary objects combined in a new arrangement that itself becomes both distinct and distinctive.

As a third level of boundary work, mainstream actors consolidate these variations as adjacent markets. Kjellberg and Olson (2017) consider the conundrum of analyzing markets that share a boundary object, applying the notion of ‘adjacent’ markets to recreational and legal cannabis. Adjacent markets bring commercial opportunities for operators to branch out. For example, Vivi described how the brand ‘Starboard’ is leveraging its competence in surfboard-crafting by diversifying into paddleboards and windsurf boards. As long as the boundary object remains – the board – then the belt of peripheral variations can emerge.

Starboard [brand] does windsurfing and paddleboarding. Naish [Robbie Naish, originally a windsurfer] does kitesurfing, surfing and paddleboarding, and NSP [Neil Pryde, originally a windsurfer] does surfing and paddleboarding. Their brands have complementary sports. I guess the paddleboarding market’s a growing market and they were trying to increase their audience. – Vivi

In addition to diversifying into multiple types of boards, we note how parallel commercial industries gravitate to the new hybrid because

of its popularity. Examples include photographers and videographers drifting into the new practice, establishing magazines and media; tour operators developing activity-specific travel activities; teenagers flocking into a highly (insta)gram-able activity because the resulting photos convey prestige. All this creative consumer engagement leads to even more variations, e.g., by mixing yoga and pilates with paddleboarding.

There are parallel industries to surf. Even yoga instructors and pilates and stuff like that, heaps of people come up with surf-specific workouts to mimic the actions of popping up or keeping your balance and core strong. Or you can buy a program in Sri Lanka for a week and its yoga/surf, yoga/surf. - Lynda

5. Discussion

Returning to the question of how consumers use boundary work to shape markets, we have shown that user innovations beginning as consumption can catalyze new market formation. To conceptualize this process, we coined the notion of ‘market bifurcations,’ defined as *contingent and emergent variations, splinterings, or splittings in which a singular seemingly cohesive consumer practice drifts into multiple, partially overlapping adjacent markets*.

In boardsports, the catalyst is the inability of commercial products to operate in all maritime conditions; thus, consumers grow frustrated with their lack of ‘board time.’ As consumers then tinker with their gear and activities (Martin & Schouten, 2014), local variations emerge. However, the new arrangements emerge as a patchwork of barely coherent practices with competing solutions across locations (c.f., translocal, Diaz Ruiz, Peñaloza, et al., 2020). The tinkerers help to stabilize the variations into niches first, and then mainstream markets (Fig. 1).

We have analyzed these events of intensive change by tracking the multiple narrative origins of each market (Spring & Araujo, 2017) to conceptualize the bifurcations, starting from local variations, and diverging into adjacent markets.

5.1. Conceptualizing ‘market bifurcations’

Our investigation into how consumers produce *translocal* variations of socio-technical arrangements shows that consumers generate *market bifurcations*. To be precise, we traced how boardsports consumers redress their frustrations with the lack of ‘board time’ by tinkering with their gear to extend their time ‘out there’ in every maritime weather condition. For example, a frustrated surfer bored with the lack of waves may take up a paddle to start enjoying the calm waters. Since the surfing arrangement is situational to waves, the incorporation of a paddle responds to the lack of waves. If enough surfers subscribe to the new socio-technical arrangement in which waves are optional, but paddles are obligatory, a new adjacent consumer practice emerges in which the paddle is crucial (Fig. 2).

Previous research shows that consumers’ customization activities can catalyze the emergence of new markets (Martin & Schouten, 2014); however, the focus is market infrastructures supporting a single object. The difference identified in this paper is that markets partially overlap, and thus we theorize that boundary work constitutes and drives market-shaping.

Recent work in market-shaping has highlighted “the systematic unfolding of orchestrated collective action” (Baker & Nenonen, 2020, p. 250), emphasizing the importance of building networks of alliances to construct ‘intermediate frames’ that while stable enough to be identified, are not markets. We extend Baker and Nenonen (2020) study to show that creative consumers can catalyze multiple translocal variations, thus forging market-like frames mediating between consumption and markets (c.f., Dolbec, 2015). Without these barely coherent frames, consumers’ generative work remains locked as solely consumption. However, we know from Shah (2000) that boardsports enthusiasts have

generated multiple commercial successes, even new markets.

There is consensus in the literature on market studies that multiple markets exist (Mele et al., 2015), but we lack nuance on how to distinguish boundaries among adjacent markets. Hence, we highlight the importance of boundary work in market formation. To explain further, as consumers put together gear from multiple sports, they break the existing boundaries, and form new ones. The emerging hybrids receive labels that emphasize similarities and contrasts with adjacent practices. In Fig. 2, we illustrate the duality of boundary and signature objects. The boundary object is the common element, while the signature object is the unique representative aspect of the new socio-technical arrangement.

The main characteristic of market bifurcations is *non-linearity* (Fig. 2). In the literature, market-shaping is represented as linear. Previous research conceptualizes a pattern of change from one initial market configuration into a second, differently shaped market configuration, as if the market was made of a plastic material (Nenonen et al., 2014). Market-shaping results in a one-to-one transformation – one market configuration changes into another configuration. However, in our conceptualization, market bifurcations occur when things fall apart, current solutions fail, and shortcomings appear. Therefore, since contingencies are hard to predict and multiple breakdowns exist, we theorize how one market can splinter into multiple adjacent markets. Fig. 2 shows a non-linear approach to market-shaping, accounting for both the continuity of the originating non-shaped configuration and simultaneously, a bifurcation that catalyzes a new market.

5.2. A radial structure for adjacent markets

This study conceptualizes market-shaping through consumption-driven boundary work. Previous work by Diaz Ruiz, Peñaloza, et al. (2020) shows that consumers generate socio-technical arrangements that vary across venues (i.e., translocal); they overlap yet remain distinct and distinguishable. However, since we are concerned with market-shaping, not consumption, it is not enough to focus exclusively on consumers’ creative work. Instead, our focus is on how these variations catalyze market formation. Therefore, we build upon DeLanda’s notion of bifurcations (DeLanda, 2016) to emphasize how the singular loses coherence when it lacks structural fit, drifting into multiple variations, each addressing specific limiting conditions. Moreover, in line with previous work on boundary objects (Star & Griesemer, 1989), we conceptualize a prototypical configuration around which variations emerge.

Fig. 3 illustrates boundary work as a radial structure around a core. We place a seemingly coherent socio-technical arrangement at the center, the core, while the belt contains the variations as a ring of peripheral categories. We choose to place the person-board arrangement in the center as it changes according to the limiting conditions illustrated in Fig. 2. The center boundary object is plastic or flexible enough to have variations while remaining recognizable, but as the boundary object mutates, each socio-technical arrangement splinters, catalyzing a bifurcation.

While a newcomer may not be able to distinguish these variations appropriately, as they all seem to be boards, an expert can associate a board type with a sport – windsurf boards have universal joints, surfboards have fins, and (some) SUP-boards are inflatable. The variations are not interchangeable; each one is unique to its boardsport. As Bowker and Star (2000) discuss, the boundary object is a platform for innovation that inherits inertia, strengths, and limitations. The board-person variations are contingent to when the singular variation breaks down because the surfer lacks waves, the board does not channel water flow properly, and the local shore is too shallow for waves but has wind gusts.

Deciding what constitutes the singular core has consequences. In our data, interviewees noted that “*everything else is a variation from surfing*” (Greg). Hence, one option is to place surfing at the core and all

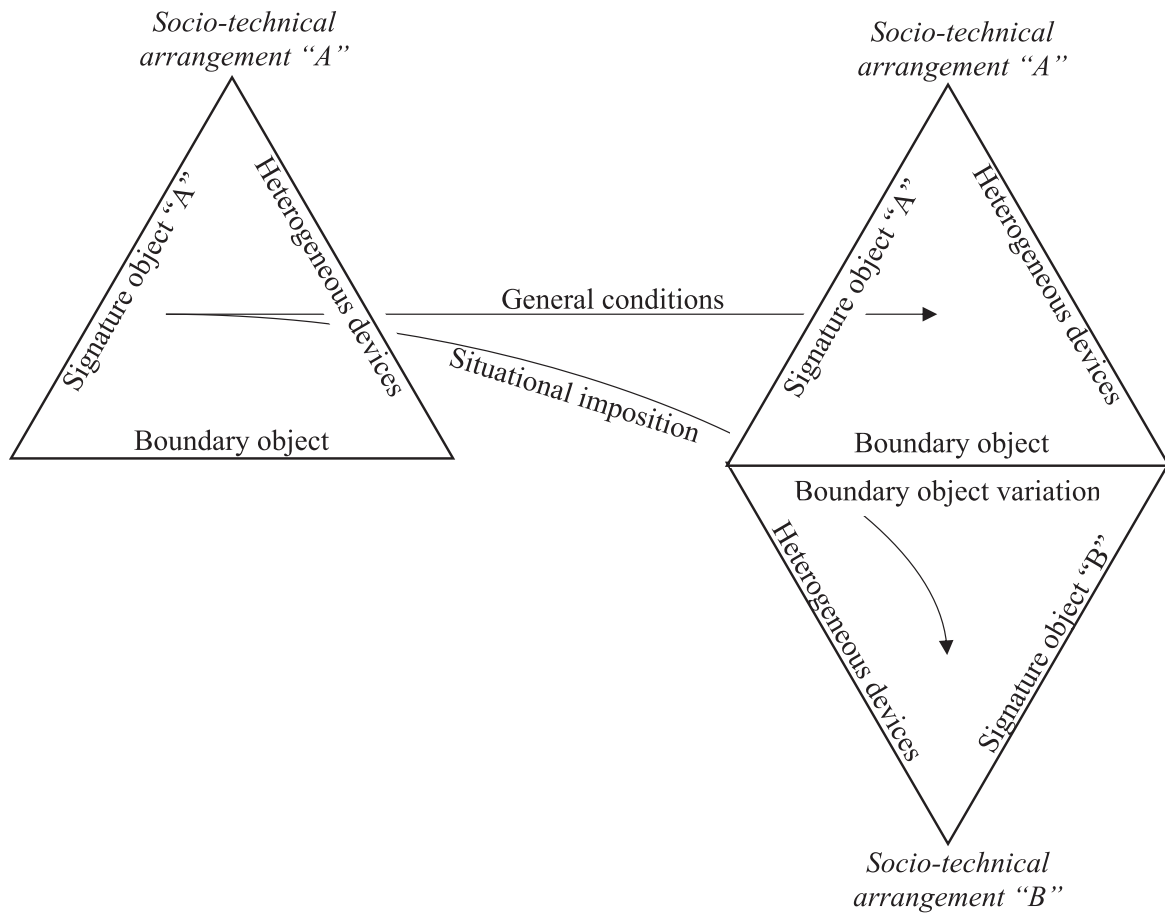


Fig. 2. Bifurcations of socio-technical arrangements.

the other boardsports around it. The corresponding implication is that every boardsport is a variation on surfing. In our interviews and also across varied data sources, 'the wave' or 'swell' is equally important. The wave as a boundary object is thus also possible – waves can be too short or too far from the shore to generate enough lift. A third option, which also emerged from the data, is to classify the variations according

to what makes them recognizable. For instance, our interviewees mentioned that boardsports are 'wind-based' or 'water-based.' This means that deciding which boundary object anchors the model is essential.

A new combination can emerge to deal with each condition by taking bits and pieces of existing practices and re-arranging them in

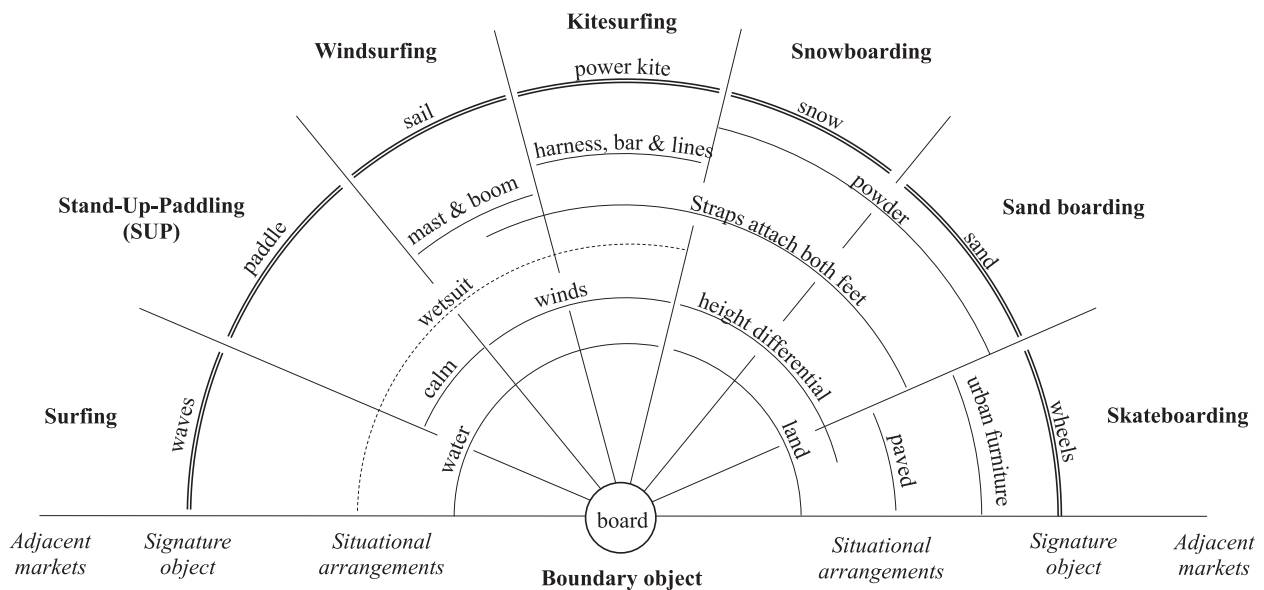


Fig. 3. The radial structure of adjacent markets.

new ways. Therefore, while all the boardsports in this paper are variations of the socio-technical arrangement *person-board-water-wind*, other variations exist that we have not studied. Some variations incorporate snow (snowboarding) or replace the ocean with asphalt (skateboarding). Therefore, the labels user-innovators give to new practices have the advantage of identifying the salient features that make them recognizable: the sail, kite, or paddle, for instance. We distinguish the salient features as per *signature objects*. By working on boundary objects, variations, and new signature objects, consumers develop cultural products that emerge to solve limiting conditions.

6. Conclusions

Through the study of user innovations in the context of boardsports, this paper shows that consumer efforts can shape markets using boundary work. We make the following three contributions.

The first contribution is conceptualizing the contingent character of consumer-driven boundary work for its potential in shaping markets. To be precise, we extend Dolbec (2015) work by fleshing out a three-step process in which first, consumers develop *translocal* variations of existing cultural products. Second, user-innovators impose coherence on the variations by creating intermediate frames. Third, the popularity of variations requires the stabilization of market practices, thus forging an adjacent market.

The second contribution applies Spring and Araujo (2017) 'product biographies' in chronically unstable markets. As pre-existing material constellations fail to fit specific local conditions, constraints emerge that impose restrictions on consumers. When commercial products break down, this study shows how consumers engage in creative consumption that shapes markets. Again, see Fig. 2. We claim that breakdowns and chronic failures in one market can lead to both the re-configuration of that market (one-to-one) and new market formation (bifurcations: one-to-many).

The third contribution explains how boundary work sets up distinctions in adjacent markets. We extend Kjellberg and Olson (2017) study by conceptualizing the boundary work of adjacent markets as per a radial structure (Fig. 3) arranged according to distinguishable (and distinct) analytical objects that we conceptualize as *signature objects*, the staples of each market.

Finally, seeing market-shaping through the prism of consumption (Giesler & Fischer, 2017; Martin & Schouten, 2014) points to novel research directions for market-shaping. First, this approach extends Harrison and Kjellberg (2016) research by considering the broader network of socio-technical arrangements surrounding market bifurcations, including consumer co-design. Second, by studying market-shaping through consumer research, opportunities emerge for understanding how consumers produce translocal variations that have the potential to become distinct consumer practices, intermediate frames, and finally, new markets. We conceptualize this process through the notion of *market bifurcations*, for which a definition is proposed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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